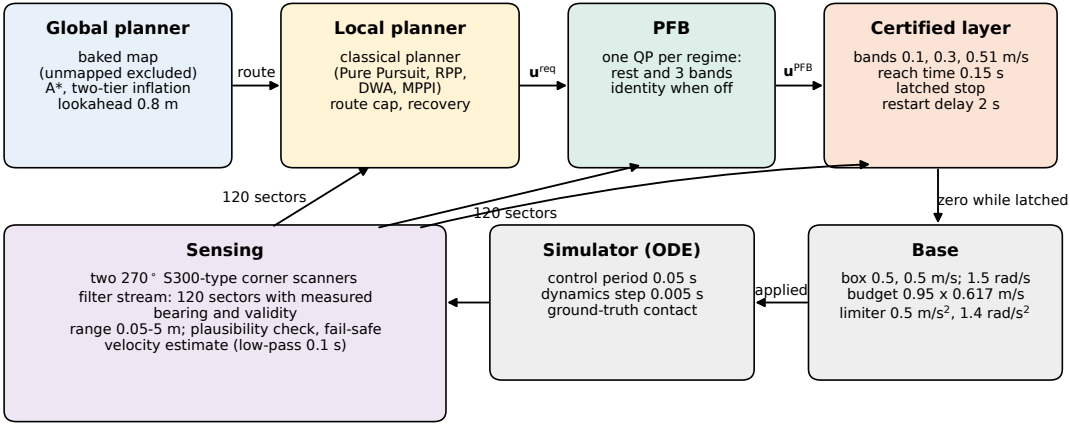
Global planner
baked map
(unmapped excluded)
A*, two-tier inflation
lookahead 0.8 m
route
Local planner
classical planner
(Pure Pursuit, RPP,
DWA, MPPI)
route cap, recovery
u^req
PFB
one QP per regime:
rest and 3 bands
identity when off
u^PFB
Certified layer
bands 0.1, 0.3, 0.51 m/s
reach time 0.15 s
latched stop
restart delay 2 s
120 sectors
120 sectors
zero while latched
Sensing
two 270° S300-type corner scanners
filter stream: 120 sectors with measured
bearing and validity
range 0.05-5 m; plausibility check, fail-safe
velocity estimate (low-pass 0.1 s)
Simulator (ODE)
control period 0.05 s
dynamics step 0.005 s
ground-truth contact
applied
Base
box 0.5, 0.5 m/s; 1.5 rad/s
budget 0.95 x 0.617 m/s
limiter 0.5 m/s², 1.4 rad/s²